

Math 196 Problem Set 5

Due April 30, at 10:00 PM

Problem 1. Suppose that A is an invertible matrix and B is a matrix similar to A . This means that there is an invertible S with $B = SAS^{-1}$. Show that B is also invertible.

Problem 2. Consider the vectors

$$v = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, w = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}$$

in the plane.

- (a) Verify that v, w form a basis for $\mathbb{R}^2$.
- (b) Compute the matrices representing projection onto $\text{span}\{v\}$ with respect to the standard basis, and with respect to the basis $\mathfrak{B} = (v, w)$.
- (c) Compute the matrices representing reflection across $\text{span}\{v\}$ with respect to these bases.
- (d) Compute the matrices representing rotation through $\pi/2$ with respect to these bases.

Problem 3. Suppose that $v_1, \dots, v_n$ is a basis for $\mathbb{R}^n$. We have some vectors $w_1, \dots, w_n$ in $\mathbb{R}^n$, and we would like to find an $n \times n$ matrix A such that $Av_1 = w_1, \dots, Av_n = w_n$.

- (a) Write these n vector equations as a single equation $AB = C$ for B, C matrices, and solve this equation to find a formula for such an A . Is this A unique?
- (b) Use your formula to find a matrix A satisfying the following equations:

$$A \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$A \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

- (c) Does any matrix A satisfy the following equations?

$$A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

Problem 4. Let V_n be the vector space of all polynomials of degree at most n in a variable t , so V_3 contains things like $2t^3 - t^2 + 5t + 1$. In this problem and the next, we will think of polynomials as inputs and outputs to functions, rather than functions themselves.

- (a) Let $M : V_1 \rightarrow V_2$ be the function $M(f(t)) = (t-1)f(t)$. Check that M is a linear transformation, and find the matrix representing it with respect to the bases $1, t$ for V_1 and $1, t, t^2$ for V_2 .
- (b) Let $S : V_3 \rightarrow V_3$ be the function $S(f(t)) = f(t-1)$, so for example

$$S(t^2 + t) = (t-1)^2 + (t-1) = t^2 - t$$

This is a linear transformation. Find the matrix representing S with respect to the basis $1, t, t^2, t^3$ for V_3 . Is this transformation invertible?

Problem 5. (Optional) In class we introduced the derivative transformation $D : V_3 \rightarrow V_3$, defined by $D(f(t)) = f'(t)$. The derivative of a degree 2 polynomial has degree 1, so we can also view D as a linear transformation $V_2 \rightarrow V_1$.

- (a) Find the matrix representation of $D : V_2 \rightarrow V_1$ with respect to the bases $1, t, t^2$ for V_2 and $1, t$ for V_1 . Using this matrix and the one you found in Problem 4 (b), find the matrix representing the linear transformation $Q : V_2 \rightarrow V_2$ defined by $Q(f(t)) = (t - 1)f'(t)$.
- (b) Find the matrix representing the linear transformation $T : V_2 \rightarrow V_2$ defined by

$$T(f(t)) = (t - 1)f'(t) - 2f(t)$$

Find a basis for the kernel of this matrix, and use it to give a description of all polynomials $f(t)$ of degree at most 2 which satisfy the differential equation:

$$(t - 1)f'(t) = 2f(t)$$

- (c) We have alternative bases $1, t - 1$ for V_1 and $1, (t - 1), (t - 1)^2$ for V_2 . Find the matrix representations of M , D , Q , and T with respect to these bases. Does this make it easier to find the kernel of T ?